

Plant Biotechnology

Lab. BT 421

Lab (8)

**Transformation of carrot slices
with *Agrobacterium tumefaciens***

By:

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Outline

- ❑ **Genetic Engineering**
- ❑ **Plant transformation**
- ❑ **Transformation methods**
- ❑ ***Agrobacterium tumefaciens***
- ❑ **The Ti plasmid and its parts**
- ❑ **The T-DNA region**
- ❑ **The mechanism of transformation**
- ❑ **Practical work**

Genetic Engineering

- ❖ The process of manipulating and transferring instructions carried by genes from one cell to another.
- ❖ Why do scientists want to change gene instructions?
 1. To produce needed chemicals.
 2. To give an organism desired characteristics of economically important crops (yield, disease resistance and abiotic stress tolerance)

Plant transformation

- ❖ **Plant transformation** is a scientific approach whereby DNA from any organism is inserted into the genome of a species of interest.
- ❖ The inserted DNA is called a “**transgene**”, and the resulting plant is said to be “**transgenic**”.

Transformation Methods

Direct Method (Widely used)

- ❖ Particle bombardment.
- ❖ Polyethylene glycol (PEG)-mediated transformation.
- ❖ Electroporation.
- ❖ Silicon carbide-WHISKERS

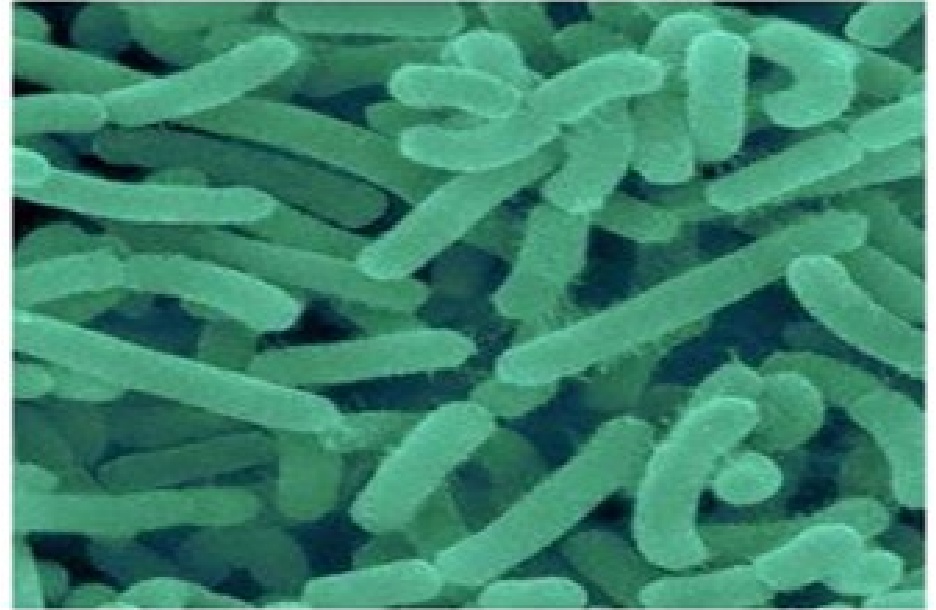
Indirect Method

- ❖ Rely on the use of *(agrobacterium)*

Agrobacterium tumefaciens

Agrobacterium tumefaciens

- ❖ Soil-borne bacteria.
- ❖ Rod shaped and gram negative bacteria.
- ❖ Has extra chromosomal DNA that contains **Ti-plasmid**.



Usage in Genetic engineering and Biotechnology

- ❖ Has the ability to infect plant cells and transfer a defined sequence of their DNA to the plant cell by infection and a causative agent of **crown gall** disease

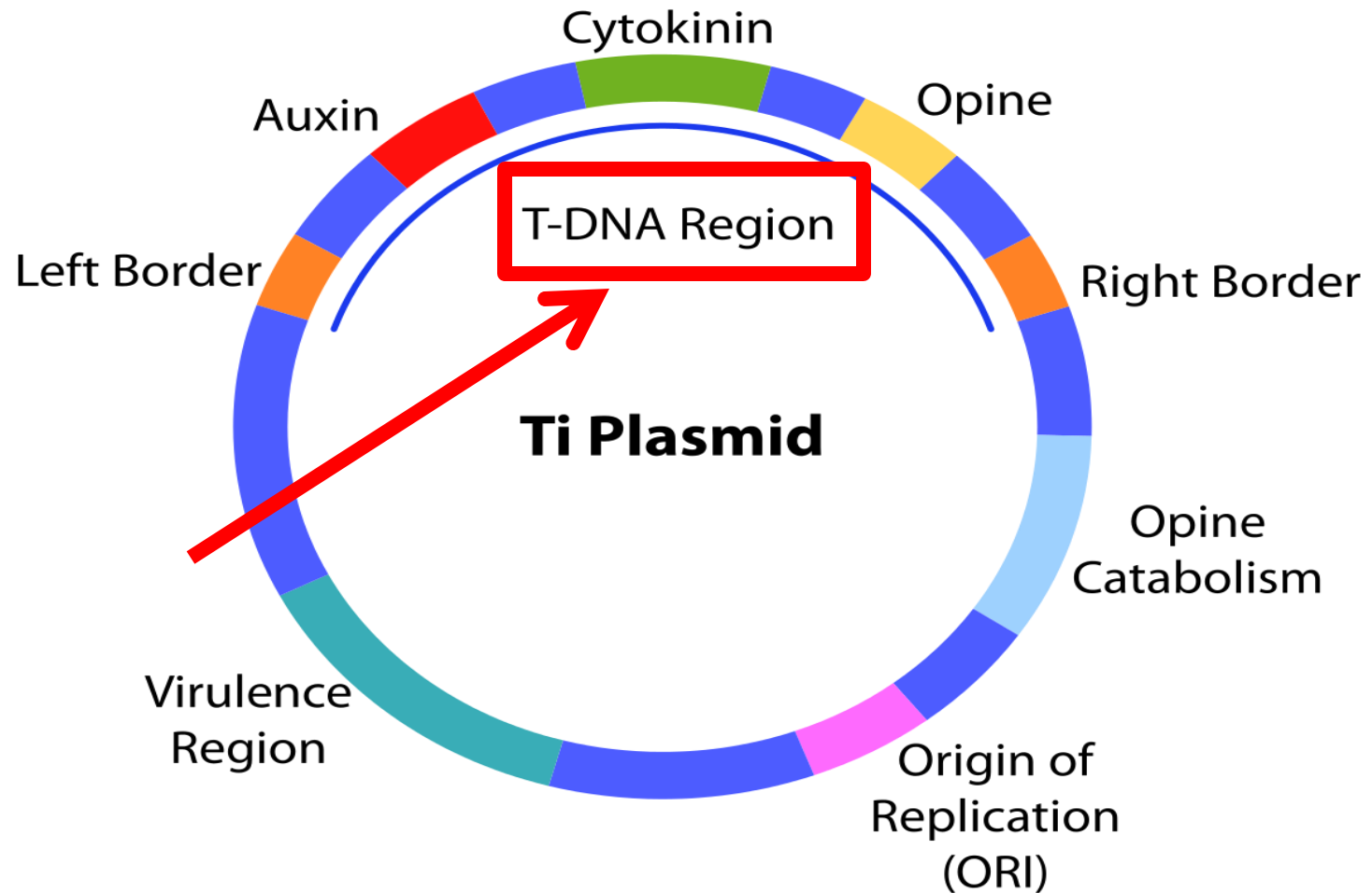
The Ti-plasmid

- ❑ A tumour inducing (Ti) **plasmid** found in **Agrobacterium** has a central role in crown gall formation.
- ❑ Ti plasmid has a specific portion called (the **T-DNA** region) that is actually integrated into the genome of the host plant.



A crown gall disease

The Ti-plasmid parts



The T-DNA region

- ❖ Any DNA between the left and right borders (24-bp) will be transferred into the genome of the host plant.
- ❖ It has **oncogenes** which are responsible for cytokinin and Auxin production.
- ❖ It has also another genes involved in the production of **opines** (a class of amino acids)

T-DNA transfer and integration steps

1. Signal recognition by Agrobacterium.

The agrobacterium detects signals (**Phenolic compounds**) released from wounded plant cells (**chemotaxis**).

2. Attachment to plant cells.

Several chromosomal **virulence genes** are involved.

3. vir Gene induction.

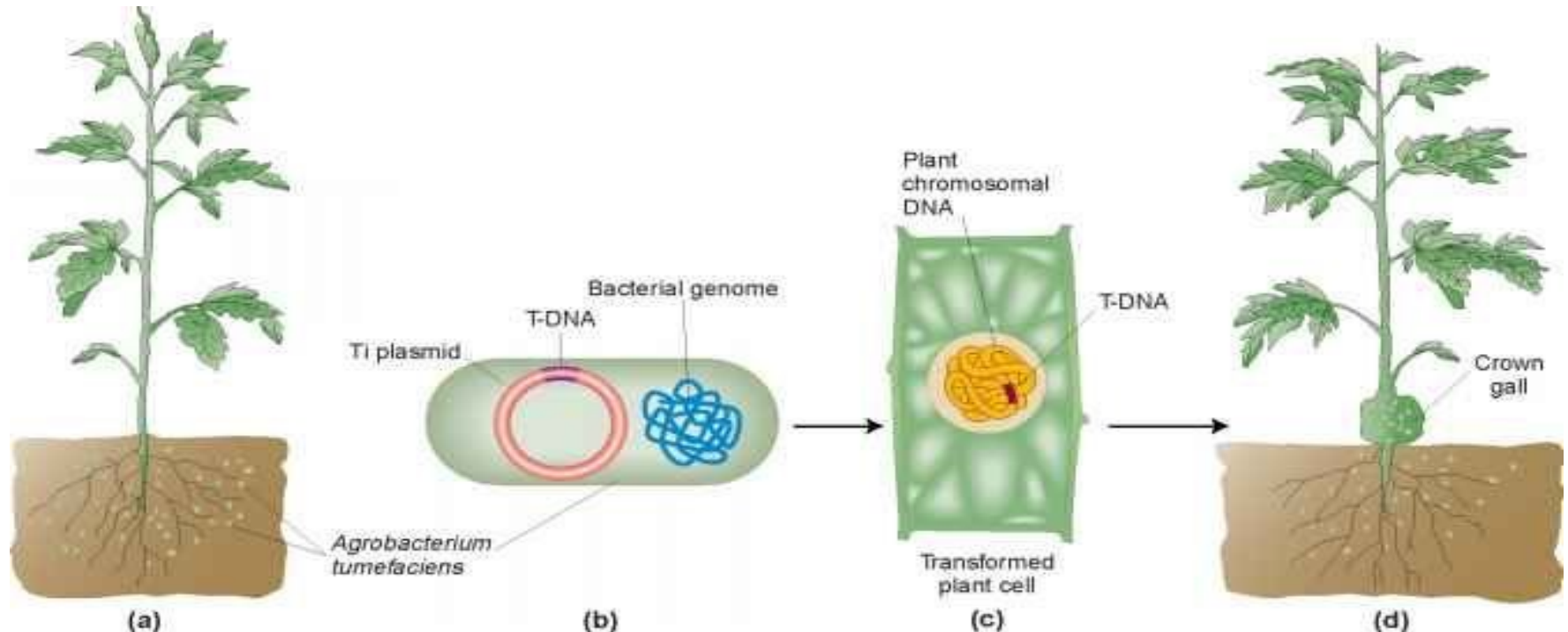
VirA (a membrane-linked sensor kinase) senses phenolics, subsequently phosphorylating and thereby activating VirG. VirG then induces expression of all vir genes.

4. T-strand production.

T-DNA transfer and integration steps

5. Transfer of T-DNA out of the bacterial cell.

6. Transfer of the T-DNA and Vir proteins into the plant cell and nuclear localisation.



The practical work

- 1. Wash a carrot with tap water to remove soil or any dust.**
- 2. Cut it into thick slices to fit the height of the petri dishes.**



The practical work

3. Sterilize the surface of carrot slices according to the next steps:

A. 10 minutes (running tap water).

B. 5 minutes (Golden soap).

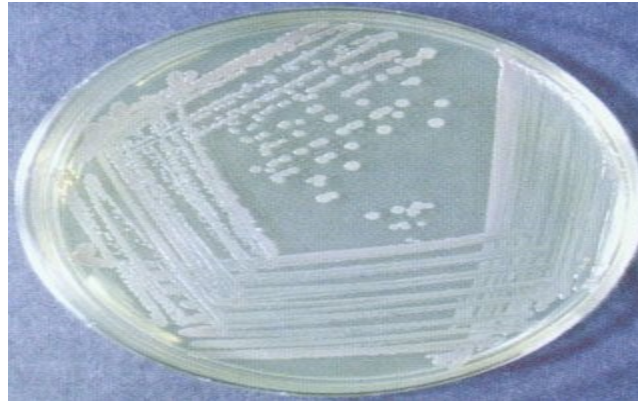
C. 10 minutes (10% Sodium Hypochlorite “Hypex”).

D. Wash the slices three times with autoclaved distilled water.

The practical work

4. Prepare your bench by wiping it with 70% alcohol and arranging the tools that you are going to use next to the spirit lamp:

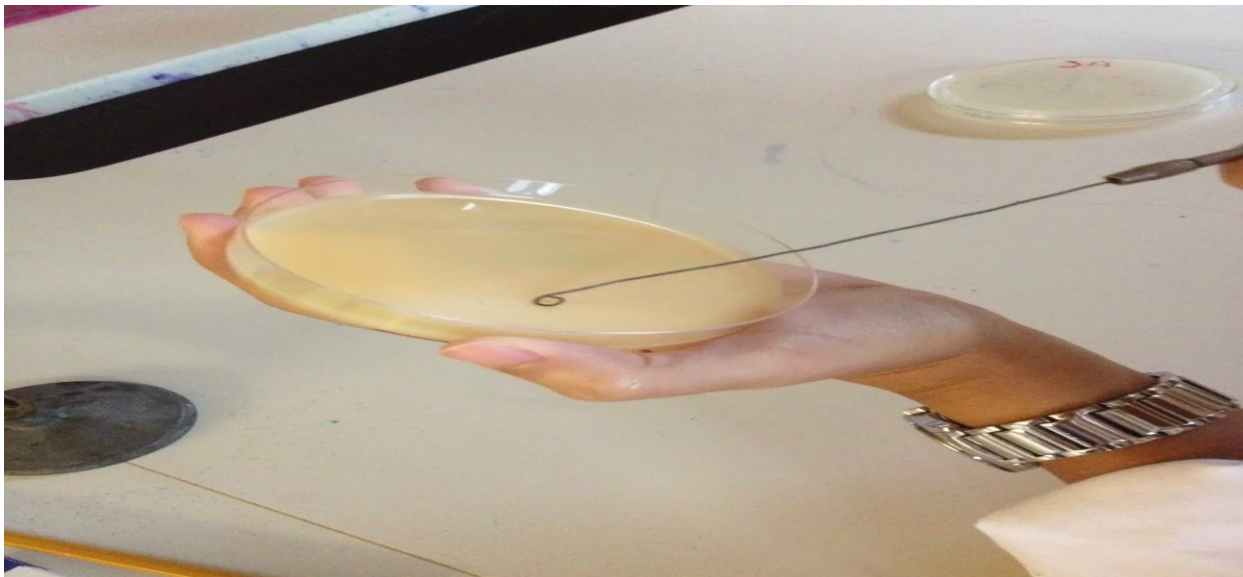
(Sterile carrot slices, a loop, 24-Agrobacterium bacterial culture, two large petri dishes lined to sterile wet filter papers).



The practical work

5. Heat the loop using the spirit lamp to sterilize it then cool it at the edge of the bacterial culture.

6. Take a loop full from bacterial plate and spread it over the first slice then cover it.



The practical work

7. Place another carrot slice without adding bacteria (Control) on the second petri dish and cover it.

8. Incubate at 27-28°C to observe the crown gall for 2 weeks.



After 2 weeks

Crown gall formation



The treated slice

The control slice

These are 2 helpful links to understand the transformation process more, please watch them.

<https://www.youtube.com/watch?v=yesNHd9h8k0>

<https://www.youtube.com/watch?v=K1ZyzvsHhOE>